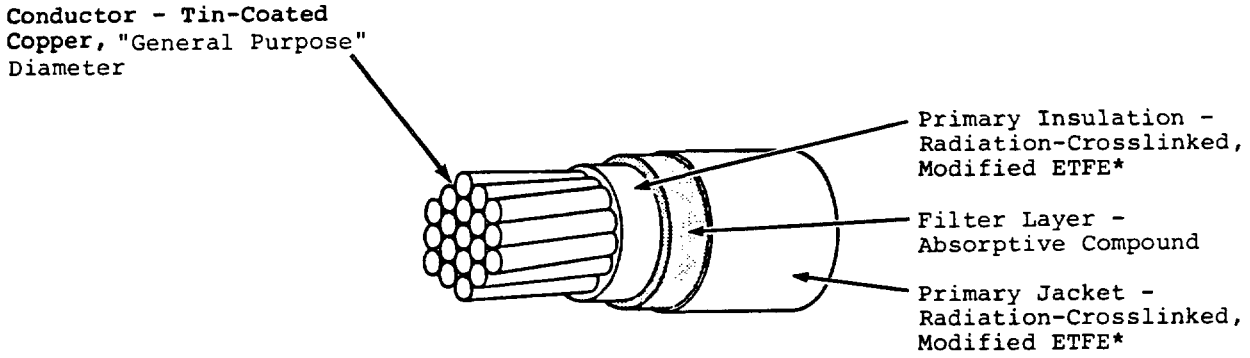
MILITARY SPECIFICATION SHEET

CABLE, ELECTRIC, FILTER LINE, COMPONENT WIRE, TIN-COATED COPPER CONDUCTOR, RADIO FREQUENCY ABSORPTIVE, 150°C, 600-VOLT

This specification sheet is approved for use by all departments and agencies of the Department of Defense.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-C-85485.



*ETFE - Ethylene-Tetrafluoroethylene Copolymer

FIGURE 1. General configuration.

TABLE I. Construction details.

Part no. 1/	Wire size	Stranding (number of strands X AWG gage of strands)	Diameter of stranded conductor (inches)		Finished wire		
			(min)	(max)	Resistance at 20°C (68°F) (ohms/1000 ft) (max)	Diameter (inches)	Weight (lbs/1000 ft) (max)
M85485/5-22-*	22	19 x 34	.029	.033	16.2	.062 ± .004	5.1
M85485/5-20-*	20	19 x 32	.037	.041	9.88	.070 ± .004	7.0
M85485/5-18-*	18	19 x 30	.046	.051	6.23	.080 ± .004	9.7
M85485/5-16-*	16	19 x 29	.052	.058	4.81	.086 ± .004	11.7
M85485/5-14-*	14	19 x 27	.065	.073	3.06	.100 ± .004	16.6
M85485/5-12-*	12	37 x 28	.084	.090	2.02	.127 ± .004	26.1
M85485/5-10-*	10	37 x 26	.106	.114	1.26	.151 ± .005	39.1

1/ PART NO.: The asterisks in the part number column, Tables I, II, and III, shall be replaced by color code designators in accordance with MIL-C-85485.

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TABLE II. Performance details.

Part no.	Bend testing			
	Mandrel diameter (inches) (±3%)		Test load (lbs) (±3%)	
	Crosslinking proof, immersion and life cycle tests	Cold bend test	Crosslinking proof, immersion and life cycle tests	Cold bend test
M85485/5-22-*	.500	1.00	.75	3.00
M85485/5-20-*	.750	1.00	1.25	4.00
M85485/5-18-*	1.00	1.50	1.75	4.00
M85485/5-16-*	1.00	1.50	2.00	5.00
M85485/5-14-*	1.00	1.50	2.00	5.00
M85485/5-12-*	1.50	3.00	2.50	5.00
M85485/5-10-*	1.50	3.00	2.50	5.00

Ratings:

TEMPERATURE RATING: 150°C (302°F) max continuous conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level

Additional requirements:

ATTENUATION (INSERTION LOSS): The wire of this specification sheet, when shielded in accordance with M85485/8-**T1*, shall meet the attenuation requirements of Table III as specified for the applicable size of wire.
BLOCKING: 200° ±3°C (392° ±5°F) for 6 hours
COLOR: Primary insulation - white
Filter layer - black
Primary jacket - light violet
COLOR STRIPING DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
CONCENTRICITY: Primary Insulation - 50% (min); Finished Wire - 70% (min)
CROSSLINKING PROOF TEST: 7 hours at 300° ±3°C (572° ±5°F)
FLAMMABILITY: 3 seconds (max); 3 inches (max)
HUMIDITY RESISTANCE: Insulation Resistance, 5000 megohms for 1000 feet (min)
IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
IMMERSION: Diameter increase 5% (max); no cracking, no dielectric breakdown
INSULATION ELONGATION AND TENSILE STRENGTH: Primary insulation, filter layer and primary jacket pulled together
Elongation, 50% (min)
Tensile Strength, 3000 lb f/in^2 (min)
INSULATION FLAWS: Primary Insulation only
Spark Test; 2.5 kilovolts (rms), 60 Hz
Impulse Dielectric Test; 6.0 kilovolts (peak)
Finished Wire
Spark Test; 3.0 kilovolts (rms), 60 Hz
Impulse Dielectric Test; 8.0 kilovolts (peak)
INSULATION RESISTANCE: 5000 megohms for 1000 feet (min)
INSULATION THICKNESS: Primary Insulation; 0.003 inch (min)
Filter Layer; average 0.003 inch (min)
Primary Jacket; 0.0035 inch (min)
LIFE CYCLE: 168 hours at 200° ±3°C (392° ±5°F)
LOW TEMPERATURE-COLD BEND: -65° ±2°C (-85° ±4°F) for 4 hours
SHRINKAGE: 0.125 inch (max) at 200° ±3°C (392° ±5°F)
SMOKE: 200° ±2°C (392° ±4°F), no visible smoke
SURFACE RESISTANCE: 500 megohm - inches (min), initial and final readings
THERMAL SHOCK RESISTANCE: Oven temperature, 150° ±3°C (302° ±5°F)
Max change in measurements, 0.060 inch
THERMAL STABILITY: 1250 hours at 180° ±3°C (356° ±5°F). The wire of this specification sheet, when shielded in accordance with M85485/8-**T1*, shall meet the voltage withstand (dielectric) of 1500 volts (rms) 60 Hz and the stop band attenuation requirements for the applicable size of wire as specified in Table III.
VOLTAGE WITHSTAND (POST-ENVIRONMENTAL): 1500 volts (rms) 60 Hz
WRAP TEST: 6 hours at 200° ±3°C (392° ±5°F); no cracking

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TABLE III. Attenuation (insertion loss).

Part no.	Pass band (dB/ft)			Transition band (dB/ft)			Stop band (dB/ft)
	1.0 MHz	10.0 MHz		100 MHz	500 MHz	1000 MHz	1 to 18 GHz
	(max)	(min)	(max)	(min)	(min)	(min)	(min)
M85485/5-22-*	.015	.04	0.10	1.3	12	30	30
M85485/5-20-*	.015	.04	0.10	1.3	12	30	30
M85485/5-18-*	.015	.04	0.10	1.3	13	33	33
M85485/5-16-*	.015	.04	0.10	1.3	13	33	33
M85485/5-14-*	.015	.04	0.10	1.3	13	33	33
M85485/5-12-*	.015	.02	0.10	1.3	12	30	30
M85485/5-10-*	.015	.02	0.10	1.3	12	30	30

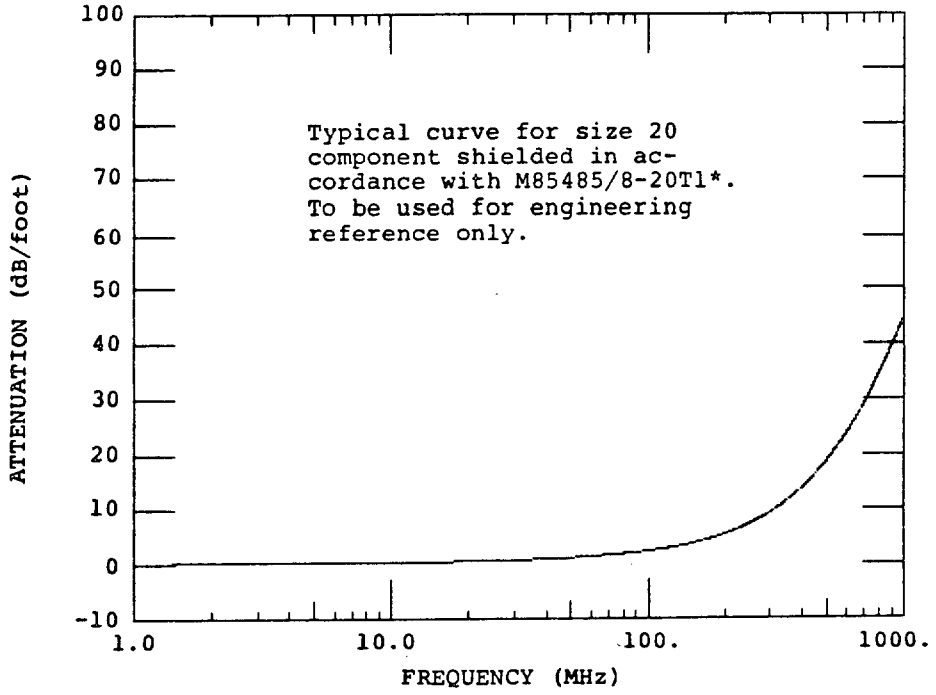


FIGURE 2. Attenuation (insertion loss).
ENGINEERING REFERENCE CURVE

Application Notes:

Component wire is an effective radio frequency absorptive wire only when used as a component in a shielded cable.

M85485/5 component wire is compatible with all standard connector and contact types, except that the primary jacket diameter of M85485/5-22 component wire exceeds the allowable diameter for MIL-C-38999 size 22D contacts. However, this component wire can be terminated to size 22D contacts if the primary jacket and filter layer are stripped in the grommet area, leaving the primary insulation intact.

Qualifying Activity: The activity responsible for this qualified products lists is the Naval Air Systems Command, Washington, D.C. 20361.

Custodian:
AF - 85

Review activity:
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Preparing activity:
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